

Section A: Multiple Choice (25 points)

All questions in this section are worth 1 point

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|----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| 1. B | 6. D | 11. D | 16. D | 21. C |
| 2. B | 7. D | 12. B | 17. A | 22. A |
| 3. B | 8. A | 13. A | 18. D | 23. D |
| 4. B | 9. B | 14. C | 19. B | 24. A |
| 5. B | 10. A | 15. A | 20. C | 25. A |

Section B: DSOs (30 points)

Questions with no point values explicitly denoted are worth 1 point.

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|---|---|
| 1. $1.34 \cdot 10^{25}$ kg | (e) WD 1856+534 belongs to a hierarchical triple-star system |
| 2. GJ 1214 b // Gliese 1214 b // Enaiposha | 10. (a) Image D |
| 3. TESS // Transiting Exoplanet Survey Satellite | (b) Tarantula Nebula |
| 4. 7 billion years old | (c) Magnetic fields |
| 5. Cancer | (d) Regulate gas motion and hold the cloud's structure intact (1), prevent gravity from taking over and collapsing the cloud into stars (1) |
| 6. Infrared (1) // Near-infrared (2) // 750-2500 nm (2) | 11. (a) Image H |
| 7. Retrograde orbit (2) | (b) Messier 42 // M42 // NGC 1976 |
| 8. First extrasolar planets to be discovered (1) // First pulsar planets to be discovered (1) | (c) JWST |
| 9. (a) WD1856+534 | (d) θ^2 Orionis (2) // Theta2 Ori (2) |
| (b) White dwarf | 12. (a) Epsilon Eridani |
| (c) WD1856+534 b | (b) Debris disc // circumstellar dust |
| (d) The exoplanet must have migrated inward after the host star evolved from red giant to white dwarf, else it would have been engulfed | (c) Poynting–Robertson effect (2) |

Section C: Free Response (45 points)

Questions with no point values explicitly denoted are worth 1 point.

- Speed of light // $3 \cdot 10^8$ m/s
- Maximum mass of a star become a stable white dwarf // 1.4 solar masses
- Its mass
- Stellar nursery // molecular cloud

5. Radial velocity method
6. Transit method (1), 70% (1)
7. Accretion or collisions (1), radioactive decay (1)
8. Tidal force // stretch and compression of the planet
9. As not to disrupt the formation of Earth-sized bodies.
10. The second (1), It is a gravitationally stable point (1), It is close enough to readily communicate with Earth (1), It keeps the Sun, Earth, and Moon behind the spacecraft for solar power (1), Provides a clear view of deep space (1), To reduce fuel consumption for course correction (1).
MAX 4 PTS.
11. Universal gravitation equation (1), density of neutron star = 10^{17} (1), 400 N (1).
12. Formulas $r_{out} = \sqrt{\frac{L}{0.53}}$ and $r_{in} = \sqrt{\frac{L}{1.1}}$ used (1). $R_{out} = 31.45$ AU (1), $R_{in} = 21.83$ AU (1), $R_{middle} = 26.64$ AU (1)
13.
 - (a) $T_{dur} = \frac{P}{\pi} \arcsin \left(\frac{\sqrt{(R_{star} + R_{planet})^2 - (bR_{star})^2}}{a} \right)$ used (1). 1.95 days (1).
 - (b) Using either $v = \sqrt{\frac{GM}{R}}$ or $v = \frac{2\pi a}{T}$ (1), final answer 1457.3 km/s (1).
 - (c) $T_{eq} = \sqrt[4]{\frac{(R_{star})^2 \cdot (T_{star})^2 \cdot (1 - \alpha)}{4D^2}}$ used (1). $T_{eq} = 411$ K (1).
 - (d) Yes, the water would boil as temperature is greater than 373K.
 - (e) +z / -z (0.5) // out of the page (1).
14.
 - (a) Using Wien's law: $T = \frac{b}{\lambda_{peak}} = \frac{0.0029}{364.6 \cdot 10^{-9}}$ (1), final answer ≈ 7950 K (1).
 - (b) F-type
 - (c) $\frac{\lambda_o}{\lambda_e} - 1 = \frac{v}{c}$, $v = 3 \cdot 10^8 \left(\frac{372.95}{364.6} - 1 \right)$ (1), final answer ≈ 6950 km/s (1).
 - (d) Using Hubble's law: $v = H_0 d$, (1), final answer 99.82 Mpc (1).
15. Title (1), x & y axes titled (0.5) correctly (0.5), units on x (0.5) & y (0.5) axes are correct, white dwarfs (1), giants (1), supergiants (1), sun located correctly (1) and sun classified as G2V (1).
Additionally, this question is a tiebreaker. Ties will be broken based on the quality of this diagram.

